Highlights

Leveraging Big Data and Cloud Technology for Scalable and Interoperable Smart Farming[⋆]

Amine Roukh, Saïd Mahmoudi

- Developed WALLeSmart, a cloud-based smart farming platform, for Wallonia region in Belgium.
- Introduced a scalable Lambda architecture integrating real-time and batch processing, achieving a processing latency of under 80 seconds for weather data streams.
- Implemented an extensible plugin system enabling tailored decision support applications, such as weather-dairy data fusion, enhancing farm efficiency and sustainability.
- Demonstrated scalability and robustness, processing over 3 million weather records and 61,130 dairy measurements with a user-friendly interface.
- Addressed data privacy and interoperability challenges with a private cloud infrastructure, supporting secure integration with external platforms and GDPR compliance.

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Leveraging Big Data and Cloud Technology for Scalable and Interoperable Smart Farming

Amine Roukh*, Saïd Mahmoudi

[a]*Faculty of Engineering - ILIA / Infortech, University of Mons, Mons, 7000, Belgium*

Abstract

Agriculture faces escalating demands to increase food production amid shrinking arable land, resource depletion, and climate variability. Existing smart farming solutions often lack scalability, interoperability, real-time analytics, and adaptability to diverse needs, limiting their impact. This paper presents *WALLeSmart*, a scalable, cloud-based smart farming management system designed to enhance interoperability while empowering agricultural stakeholders with data-driven decision-making tools. A standout feature of the platform is its modular plugin architecture, which enables agricultural professionals and developers to build custom applications tailored to specific farming needs. These applications leverage the platform's comprehensive data ecosystem and interoperable infrastructure to integrate diverse data sources—such as IoT sensors, weather data, and satellite imagery—enabling precision irrigation, optimized fertilization, and targeted pest control. Built on a scalable cloud infrastructure, *WALLeSmart* supports a wide range of users, from smallholder farms to large agricultural enterprises. The paper also addresses critical challenges in data security, privacy, and cross-platform compatibility, proposing robust frameworks to enable secure and interoperable data exchange. Through real-world case studies and pilot implementations, we demonstrate how *WALLeSmart* fosters an interconnected agricultural landscape, enhancing efficiency, sustainability, and profitability. As a use case, *WALLeSmart* aggregates and processes over 3 million historical weather measurements and 61,130 dairy farm datasets from the Wallonia region of Belgium to deliver real-time insights and predictive analytics, showcasing its ability to integrate and analyze diverse agricultural data sources effectively.

Keywords: Smart Farming, Big Data, Cloud Computing, IoT, Precision Agriculture, Predictive Analytics, Dairy Farming, Decision Support Systems, Interoperability

1. Introduction

The global population is projected to grow significantly over the next several decades, reaching a peak of approximately 10.3 billion people by the mid-2080s, up from 8.2 billion in 2024 [2]. This unprecedented growth poses immense challenges for agriculture, including the need to substantially increase food production, ensure sustainable livelihoods for farmers, and protect environmental health. These challenges are further exacerbated by the shrinking availability of arable land due to urbanization and environmental degradation, the depletion of critical natural resources such as freshwater [3], and the increasingly unpredictable impacts of climate change, including extreme weather events and shifting growing seasons [4]. In this context, the agricultural sector must adopt innovative solutions to enhance productivity, sustainability, and resilience.

Precision agriculture and smart farming have emerged as transformative approaches to address these challenges. Precision agriculture leverages technologies such as global navigation satellite system (GNSS), sensors, and data analytics to optimize field-level management with remarkable accuracy [5]. Building on these principles, smart farming integrates advanced technologies like the Internet of Things (IoT), drones, and artificial intelligence (AI) to create a more comprehensive and adaptive agricultural management system [6]. By incorporating real-time data, situational awareness, and location-specific insights, smart farming enables farmers to optimize resource use, predict potential issues, and improve overall productivity [7]. Moreover, these technologies promote sustainable practices, such as precise water and nutrient management, which reduce waste and minimize environmental impact [8]. Early detection of pests and diseases through real-time monitoring and predictive analytics further reduces reliance on chemical treatments, enhancing both crop health and environmental sustainability [9].

The rapid advancement of IoT and cloud computing technologies is driving a paradigm shift in agriculture. IoT-based solutions, such as livestock and soil sensors, are becoming increasingly prevalent, providing farmers with critical data to optimize their operations [10]. For instance,

*This work is an extension of our paper [1]

*Corresponding author

Email addresses: amine.roukh@umons.ac.be (Amine Roukh), said.mahmoudi@umons.ac.be (Saïd Mahmoudi)

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livestock sensors enable ranchers to monitor the health, location, and behavior of animals in real-time, improving animal welfare and optimizing feed and breeding programs [11]. Soil sensors offer detailed insights into topography, moisture levels, and nutrient content, enabling precise irrigation and fertilization while reducing water and fertilizer waste [12]. These sensors also provide localized weather forecasts, helping farmers make informed decisions about planting, harvesting, and risk management. Additionally, self-driving tractors and agricultural drones are revolutionizing labor-intensive tasks. Remote-controlled tractors significantly reduce labor costs [13], while drones can reduce operation costs and an increase productivity by 83.3% in manpower for herbicide application and 41.6% when used for fertilizing [14].

Digital technologies, including smartphone applications and cloud-based platforms, are further empowering farmers to monitor and manage their operations remotely. These tools enable real-time tracking of equipment, crops, and livestock, as well as access to actionable insights through data analytics [15]. Cloud platforms facilitate the integration of historical data with predictive analytics, allowing farmers to anticipate market trends, weather fluctuations, and potential challenges [16]. However, the widespread adoption of these technologies generates vast amounts of data. IBM estimates that the average farm can produce up to half a million data points per day [17]. While big data holds immense potential for transforming agriculture—enabling predictive analytics, proactive decision-making, and full automation of the agri-food chain—it also introduces significant challenges. Managing this data, from acquisition and processing to storage, analysis, and visualization, requires sophisticated architectural frameworks and tools [18]. Key challenges include handling large, heterogeneous datasets from diverse sources, processing data in real-time and batch modes, applying advanced analytical techniques, and presenting insights through user-friendly interfaces.

Looking ahead, the integration of IoT with Artificial Intelligence (AI) and Machine Learning (ML) holds even greater potential for transforming the agricultural landscape. AI-driven predictive models can analyze vast amounts of data collected from sensors and drones, offering advanced insights into crop growth patterns, disease risks, and optimal harvesting times [19]. Machine learning algorithms can also continuously improve these models based on new data, allowing for increasingly precise recommendations that help maximize yield and minimize resource use. Cloud computing further enhances this by offering scalable, on-demand data storage and processing power, enabling farmers of all sizes to adopt these cutting-edge technologies without needing significant in-house infrastructure [20].

The integration of AI and cloud computing into agriculture faces several challenges, including data privacy and security concerns due to the sensitive nature of agricultural data, interoperability issues arising from diverse hardware and software systems, and data quality and management hurdles that impact the reliability of AI-driven insights [21]. Additionally, connectivity and infrastructure limitations in rural areas, high costs of implementation, and a lack of technical expertise among farmers hinder widespread adoption. Ethical concerns, such as algorithmic bias and data ownership, as well as the need for scalability and adaptability across diverse farming contexts, further complicate integration [22].

To address these challenges, this paper introduces *WALLeSmart*[1], a scalable, cloud-based smart farming platform designed to help farmers, researchers, and administrators optimize agricultural operations and decision-making. The platform integrates IoT sensors, satellite imagery, weather and milk data to provide real-time insights and predictive analytics. A key feature of *WALLeSmart* is its extensible plugin system, which allows developers and agricultural professionals to create custom applications for specific farming needs, such as optimized irrigation, fertilization, and pest control. By addressing critical challenges such as data privacy, security, and interoperability, *WALLeSmart* offers a robust and user-friendly solution for modern agriculture.

1.1. Our Contributions

The key contributions of this paper are:

- A detailed survey of existing smart farming platforms, with a comprehensive review of state-of-the-art big data processing architectures and cloud computing solutions for agriculture;
- A novel scalable and interoperable architecture that integrates real-time and batch processing for agricultural data management;
- An extensible plugin system that facilitates seamless integration of custom decision support applications, enhancing interoperability across diverse farming technologies;
- Implementation and validation of the system using real-world data from 30 dairy farms and 32 weather stations in Wallonia, showcasing its ability to process heterogeneous agricultural datasets;
- The development of an interactive web application for data visualization and decision support, ensuring accessibility and usability for various agricultural stakeholders;
- The integration of IoT devices, geospatial data, and predictive analytics to enable data-driven precision agriculture practices;
- An empirical evaluation demonstrating system performance, scalability, and interoperability under various workloads and data sources.

[1] *WALLeSmart* can be found at: `https://wallesmart.be`

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1.2. Paper Organization

The remainder of this paper is organized as follows. Section 2 reviews existing smart farming platforms and tools. Section 3 provides a background on smart farming technologies. Section 4 presents the design and components of the *WALLeSmart* platform. Section 5 details the implementation and evaluation of the platform using real-world data. Finally, Section 6 summarizes the key findings and outlines future research directions.

2. Related Work

The rapid evolution of smart farming technologies has led to the development of numerous platforms, tools, and methodologies aimed at optimizing agricultural practices. This section provides a comprehensive review of the most relevant works in the field, focusing on their contributions, limitations, and how they relate to the proposed *WALLeSmart* platform.

Cloud computing has become a cornerstone of modern smart farming, enabling scalable data storage, processing, and analysis. *FarmBeats*, developed by Microsoft, integrates IoT sensors, drones, and cloud computing to provide farmers with real-time data on soil conditions, weather, and crop health [23]. While *FarmBeats* showcases the potential of IoT in agriculture, challenges such as technology adoption barriers and the need for farmer training remain critical for widespread implementation. Similarly, [24] paper proposes an agri-cloud framework to improve the cloud computing framework of agriculture data as a service (ADaaS) over the public cloud. The model provides accurate and secure data sharing to the various stakeholders, using a security attribute-based group signature (ABGS) system. The paper of [25] introduces a Farm Management System (FMS) framework for service interoperation. The FMS allows for a marketplace of services that can share data securely. A proof of concept was developed under the *SmartAgrifood Project* and tested in a greenhouse in Crete, Greece, over nine months to identify technical issues. The FMS handles various types of data, and the complexities surrounding copyright and ownership need to be addressed. Moreover, while the paper discusses the creation of a marketplace for services, the integration of various services from different providers can be complex. Ensuring seamless interoperability among diverse applications may present technical challenges that could hinder the system's usability and effectiveness.

The applications of big data in the agriculture field are numerous, we can cite for instance: precision farming, yield prediction, risk mitigation, loss reduction, supply chain management, farm-to-fork traceability, and sustainable farming [34, 35, 6, 36]. Besides, agricultural big data systems can be broadly divided into three categories: (i) advanced sensor technology systems, (ii) risk management systems, and (iii) agricultural management systems [37]. Recently, initiatives are ongoing to create agricultural platforms that collect the data needed by smart farming decision support tools. They are often created by private companies or public-private partnerships [38]. Among the former, Monsanto's *Integrated Farming Systems* platform collects data on information such as soil health and pest pressures and provide them for farmers. The Climate Corporation proposes the *Climate FieldView* platform to aggregate data of different sources in one place and provides diagnostic and applicative tools to farmers [26]. However, since 2018 only four companies dominate the market: Corteva Agriscience, Syngenta Group, BASF SE, and Bayer AG due to a series of mergers and acquisitions [39]. Regards public-privates partnerships, many countries are building systems to advance the use of modern agricultural technologies. For example, *farmmaps* (previously Akkerweb) is a Dutch web-based platform that provides access to external data sources such as weather, parcel boundaries, satellite, and data from commercials farm management system. It stores geo-referenced data, including soil maps and drone imagery. It allows combining data sources and processing of data, through a set of application modules to provide farmers decision support and recommendations [27]. Another example is the platform *Barto* in Switzerland [28]. As a stock company, *Barto* brings together public and private actors to build up a smart-farming platform that also aims to digitize operational and production farms data while avoiding duplication, which speeds up on-farm processes, reduces administrative tasks. *Barto* itself is based on *365FarmNet*, a German-based farm management software provider. Nearly 45.000 farmers are already active on this platform in Germany, Poland, Bulgaria, Austria, and France [2]. The platform operates as SaaS (Software as a service) and provides solutions for managing and recording all activities on a farm. However, those platforms are proprietary, and therefore the internal system architecture is not available. In the academic sector, many kinds of researches have been conducted in the smart farming field. The authors of [30] from Spain, demonstrate the advantage of a tool, called *PLATEM*, that applies real-time decisions from data such as variable rate irrigation, and selected parameters from field and weather conditions. Data is processed in a decision-making system based on learning prediction rules. *PLATEM* is a collaborative tool where farmers can post their experiences. It can monitor the different parameters of soil and environment to better the growth of crops. Those tasks can be accomplished through a chain of web-based graphical tools. Nevertheless, data collection is not detailed and MySQL database management is used to store it. Similarly, *CropSense* uses machine learning algorithms to analyze historical and real-time data, providing recommendations for planting and harvesting [29]. Another initiative, *DEMETER*, focuses on addressing interoperability challenges in smart agriculture [31]. By integrating diverse hardware and software resources, *DEMETER* facilitates seamless data exchange

[2] `https://www.365farmnet.com`

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Table 1: Comparison of Smart Farming Platforms

Platform	Scalable	Interoperable	Extensible	Batch Processing	Real-time Processing	Cost-effective
FarmBeats [23]	✓	×	×	✓	✓	◐
Climate FieldView [26]	✓	◐	×	✓	✓	×
farmmaps [27]	✓	×	✓	✓	×	✓
Barto [28]	◐	✓	✓	✓	×	✓
CropSense [29]	◐	×	×	✓	×	✓
PLATEM [30]	×	×	×	✓	✓	✓
DEMETER [31]	✓	✓	✓	◐	◐	×
DSSAT [32]	×	◐	✓	✓	×	✓
CropWise [33]	✓	×	✓	✓	×	×

(✓): Supported; (×): Not supported; (◐): Partially supported.

and knowledge sharing across the agri-food chain. The platform has been tested in two trials in the Murcia region of Spain, demonstrating its benefits for arable crop management. Despite these advancements, many existing solutions continue to face challenges related to data heterogeneity, real-time processing, and user-friendly visualization—factors that are crucial for widespread adoption by farmers [18].

The work of [40] utilize IoT sensors to monitor electrical conductivity, pH, temperature, and humidity, enabling precise irrigation and fertilization. Similarly, studies of [41, 42] employ IoT devices to track animal health and behavior, improving livestock management. However, these systems often lack integration with broader data ecosystems, such as weather forecasts and satellite imagery, limiting their predictive capabilities. Recent advancements in edge computing have sought to address these limitations by enabling real-time data processing at the source, reducing latency and improving efficiency [43].

The advent of autonomous farming technologies, such as self-driving tractors and drones, has further revolutionized agriculture. The authors of [44] proposed a drone-based agricultural monitoring system using Optical Camera Communication (OCC), enabling high-density sensing at low cost, with a trajectory control algorithm reducing flight time by 30% and achieving low bit error rates in real-world experiments. Similarly, autonomous robot developed by [45] employs a CNN-based system to detect and address tomato plant diseases, autonomously activating a pesticide sprayer for precision targeting, mitigating risks associated with human error in sustainable farming practices. However, these technologies often operate in isolation, lacking integration with comprehensive farm management systems. Recent efforts have focused on developing interoperable systems that combine autonomous machinery with IoT and cloud-based platforms, as demonstrated by [46].

Cloud-based decision support systems (DSS) have emerged as powerful tools for agricultural management. Platforms like *CropWise* by Syngenta provide farmers with actionable insights based on historical and real-time data, enabling optimized planting, fertilization, and harvesting [33]. Another example is *DSSAT* (Decision Support System for Agrotechnology Transfer), a tool used for simulating crop growth and yield under various environmental and management conditions. It integrates data on soil, weather, and crop management to provide insights into agricultural practices, enhancing productivity and sustainability [32]. The DSSAT model has been utilized to compare conventional and alternate wetting and drying (AWD) techniques in rice farming, demonstrating a 99.78% accuracy in predicting grain yield and improved water use efficiency with AWD [47]. While these systems excel in data integration and visualization, they often lack extensibility, making it difficult for users to develop custom applications tailored to their specific needs. Recent advancements in predictive analytics, such as those proposed by [48], have sought to address these limitations by enabling more accurate and timely decision-making.

A comparative analysis of existing smart farming platforms is presented in Table 1, highlighting key capabilities such as scalability, interoperability, extensibility, batch and real-time processing, and cost-effectiveness. The table demonstrates that while several platforms, such as *FarmBeats* and *CropWise*, exhibit strong scalability, many solutions struggle with interoperability and extensibility. These insights emphasize the need for a more comprehensive and modular approach in smart farming solutions.

While existing smart farming platforms and tools have made significant strides in optimizing agricultural practices, several gaps remain. First, many platforms are not designed to address region-specific challenges, such as the diverse climatic conditions and agricultural practices. Second, the integration of heterogeneous data sources, including IoT sensors, satellite imagery, and weather forecasts, remains a challenge due to issues of interoperability between systems and platforms. Third, while some platforms offer predictive analytics, few provide extensible frameworks for developing custom applications. Finally, user-friendly interfaces and real-time data processing capabilities are often lacking, hindering widespread adoption.

The proposed platform addresses these gaps by offering a scalable, cloud-based solution tailored to the needs of the Wallonia region of Belgium. By integrating IoT sensors, satellite imagery, and weather data, *WALLeSmart* provides real-time insights and predictive analytics

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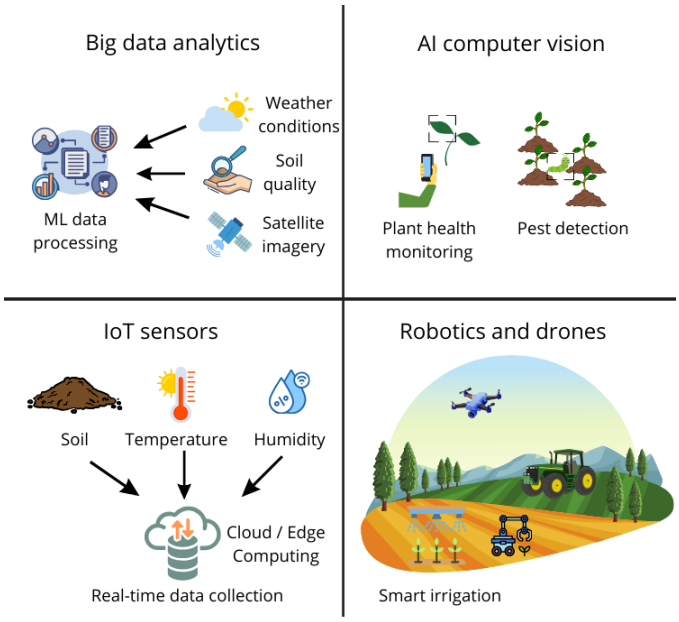

Figure 1: Overview of smart farming technologies.

to support data-driven decision-making. Its extensible plugin system allows developers and agricultural professionals to create custom applications, addressing specific farming needs such as optimized irrigation, fertilization, and pest control. Furthermore, *WALLeSmart* emphasizes user-friendly data visualization and real-time processing, making it accessible to farmers of all scales. Before detailing our proposed system, in the next section, we provide an overview of smart farming technologies and their role in transforming traditional agricultural practices.

3. Smart Farming Technologies Overview

The integration of advanced technologies into agriculture has transformed traditional farming practices, ushering in the era of smart farming. Figure 1 presents a comprehensive overview of the core components driving this transformation. IoT sensor devices play a critical role in measuring parameters such as soil quality, temperature, and humidity, offering real-time data to monitor environmental conditions. Robotics and drones are utilized for tasks like aerial imaging, crop monitoring, and precision operations, significantly enhancing efficiency and accuracy. Real-time data collected from these diverse sources empowers farmers to make timely and informed decisions. At the heart of this ecosystem lies data processing and analytics, enabling the management and analysis of large agricultural datasets to extract actionable insights. Collectively, these technologies enable seamless data collection, processing, and analysis to advance modern agriculture.

The exponential growth of data in agriculture, driven by IoT sensors, drones, satellites, and farm machinery, necessitates robust big data processing architectures and scalable cloud computing solutions, which form the backbone of smart farming. In the next sections, we will discuss the key components of big data processing architectures and the role of cloud computing in agriculture.

3.1. Big Data Processing for Smart Farming

Big data refers to both structured and unstructured data that is too large or complex to be processed using traditional data processing tools and techniques. The term "big" is characterized by the 5 Vs: Volume, Velocity, Variety, Veracity, and Value [49]. Some companies have also introduced additional dimensions to further describe big data. To fully harness the potential of this vast amount of data, it must be effectively managed and analyzed. This process begins with ingesting and integrating raw data from diverse sources into a centralized storage repository known as a Data Lake. Next, big data solutions are employed to process the data, utilizing both long-running batch jobs and real-time processing to filter, aggregate, and prepare the data for analysis. The final step involves extracting actionable insights through data analysis and reporting.

In general, big data systems must meet two fundamental requirements: (1) the ability to handle massive real-time data streams from multiple sources, and (2) the capability to perform immediate analysis to deliver timely results [50]. To address these requirements, three state-of-the-art software architectures are commonly used: *Lambda*, *Kappa*, and a *Hybrid* that combines elements of both.

3.1.1. Lambda Architecture

The Lambda architecture describes a generic, scalable, and fault-tolerant real-time data processing framework [51]. It consists of two parallel processing branches: one for batch processing and another for real-time processing. As illustrated in Figure 2, incoming data is simultaneously fed into both branches. In the batch layer, data is appended to a centralized storage area known as the master dataset. This data is typically processed at a later stage using batch processing tools like Apache Hadoop, generating batch views. In contrast, the speed layer processes data in real time using stream processing tools such as Apache Storm, producing incremental views. Since processing large datasets in the batch layer can be time-consuming, the results are often not up-to-date. The speed layer addresses this limitation by providing access to the most recent data. Finally, the serving layer merges the outputs from both the batch and speed layers, enabling end-users to query the combined data for comprehensive insights.

3.1.2. Kappa Architecture

The Kappa architecture aims to simplify the development and maintenance of data processing systems [52]. Unlike the Lambda architecture, it utilizes only one processing branch—the speed layer—as illustrated in Figure

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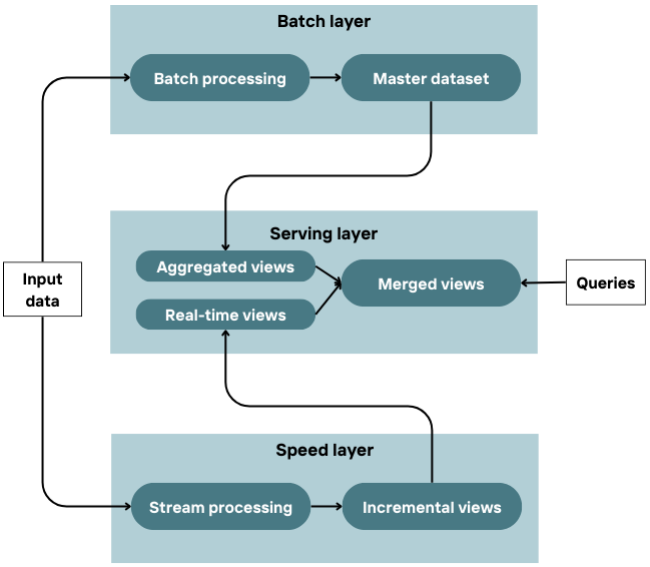

Figure 2: *Lambda* architecture.

3. In this architecture, incoming data is processed immediately in real time using a single stream processing tool, such as Apache Kafka or Apache Flink. The results are stored as incremental views, which are then made available in the serving layer for querying and analysis. A key feature of the Kappa architecture is its ability to reprocess the entire master dataset as needed, while simultaneously handling incoming real-time data streams. This dual capability ensures flexibility and scalability. The primary advantage of the Kappa architecture lies in its single data processing engine, which simplifies system design and reduces complexity. However, this approach may require increased computational power and storage capacity to handle the parallel processing of large datasets and real-time data streams efficiently.

3.1.3. Hybrid Architecture

While both the *Lambda* and the *Kappa* architectures support the processing of historical data and real-time data, they operate in separate modes. This separation requires users to manually integrate the results from both processing modes to enable a comprehensive analysis of the data [53]. To address this limitation, [53] introduced a hybrid processing architecture called *BRAID*. *BRAID* intertwines the processing of historical and real-time data by establishing communication channels between the batch engine and the stream engine. This integration allows for automatic comprehensive analyses with minimal overhead. The authors also explored various implementation techniques for their proposed architecture. In a similar vein, the study by [54] proposed a conceptual architecture for big data streaming integrated with complex event processing (CEP), named *BiDCEP*. This system extends the *Lambda* and *Kappa* architectures to accommodate the complex event processing and event management domains of enterprise IT. The authors initiated a technical discussion on the benefits of combining these architectures and provided a motivational example to illustrate their approach.

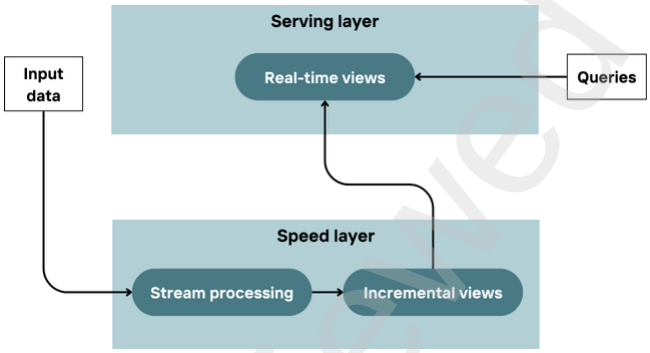

Figure 3: *Kappa* architecture.

3.2. Cloud Computing for Smart Farming

Cloud computing is a technology that enables on-demand access to shared pools of configurable computing resources (e.g., servers, storage, networks, applications, and services) over the internet. These resources can be rapidly provisioned and released with minimal management effort or service provider interaction [55]. Cloud computing is defined by five essential characteristics, three service models, and four deployment models. The key service models include Infrastructure as a Service (IaaS), which provides virtualized computing resources; Platform as a Service (PaaS), which offers development and deployment platforms; and Software as a Service (SaaS), which delivers applications over the internet. The four deployment models are Private Cloud, Community Cloud, Public Cloud, and Hybrid Cloud, each catering to different organizational needs and use cases [55]. Many agricultural businesses adopt hybrid cloud models that combine on-premise infrastructure with cloud services. This approach offers flexibility and ensures sensitive data remains securely stored locally, while benefiting from the scalability of the cloud. Cloud computing offers numerous benefits, such as scalability, cost efficiency, and flexibility making it a valuable tool for modern agriculture. However, its adoption also introduces several challenges. Key issues include security and privacy, downtime and reliability, cost management, data transfer limitations and vendor lock-in, are the main challenges in the use of cloud, especially in the agricultural domain [56].

3.2.1. Fog Computing in Smart Farming

While cloud computing provides centralized processing, fog computing extends cloud computing by acting as an intermediate layer between edge devices and the cloud. Fog computing is defined as a decentralized computing architecture that extends cloud computing capabilities to the edge of the network, closer to the data source. It enables data processing, storage, and analysis at the edge

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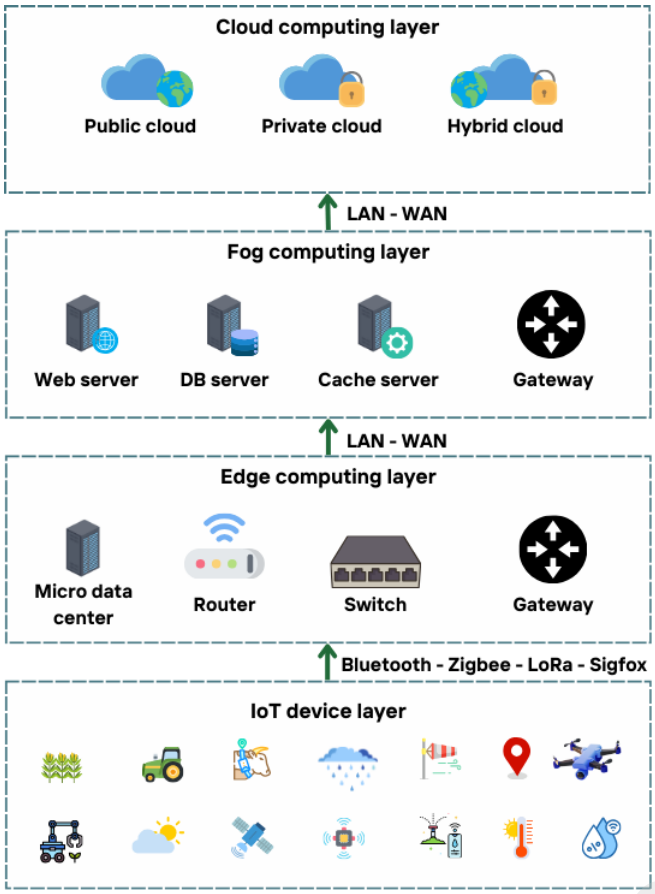

Figure 4: Overview of cloud, fog, and edge computing layers in smart farming architecture.

devices (e.g., IoT sensors, gateways) rather than sending all data to a centralized cloud. This approach reduces latency, bandwidth usage, and reliance on cloud infrastructure while supporting real-time decision-making and localized data processing [57]. Fog computing is particularly useful in applications requiring low latency, such as smart farming systems [58]. The Fog layer consists of large-scale, geo-distributed Fog nodes deployed at the edge of networks. Each Fog node is equipped with on-board computational resources, data storage capabilities, and network communication facilities, enabling it to act as a bridge between IoT devices and the cloud within the IoT ecosystem [59].

3.2.2. Edge Computing in Smart Farming

Edge computing is a distributed computing paradigm that brings data processing and storage closer to the source of data generation, such as IoT devices, sensors, or local servers, rather than relying on a centralized cloud infrastructure. By processing data at the "edge" of the network, edge computing reduces latency, minimizes bandwidth usage, and enables real-time decision-making [60]. While both Fog computing and Edge computing aim to reduce latency and improve efficiency by decentralizing data processing, they differ in their architecture, scope, and applications. Fog computing uses hierarchical layer of Fog nodes (e.g., routers, gateways) enabling distributed intelligence across a network, while Edge computing emphasizes localized processing at the data source [59]. Edge computing has numerous applications in agriculture, including precise irrigation, fertilization, pest control, livestock Monitoring, and supply chain traceability [22].

3.2.3. Serverless Computing

Serverless computing, a relatively new and rapidly evolving field in cloud computing. Serverless is a model where the cloud provider dynamically manages the allocation and provisioning of computing resources, allowing developers to focus solely on writing and deploying code without worrying about underlying infrastructure. In serverless computing, applications are broken into small, event-driven functions (e.g., AWS Lambda, Google Cloud Functions) that are executed in response to specific triggers, such as HTTP requests or database updates. The cloud provider automatically scales resources based on demand, and users are billed only for the actual execution time of the functions, making it a cost-efficient and scalable solution for modern applications [61]. Serverless architectures allow smart farming applications to run event-driven tasks without managing servers. For example, a serverless function can trigger real-time notifications when a sensor detects critical thresholds in soil moisture levels [62].

Figure 4 depicts a multi-layered architecture for smart farming, consisting of IoT devices for data collection, edge computing for local processing, fog computing for intermediate analysis, and cloud computing for advanced analytics and storage.

4. WALLeSmart System Architecture

In this section, we present the architecture of the *WALLeSmart* framework. Drawing from the previous discussions, we adopt a *Lambda* architecture to implement our system. This choice is motivated by the distinct nature of batch and real-time analysis in our context, necessitating separate tools for each task. Additionally, the *Lambda* architecture ensures linear scalability, fault tolerance against hardware failures, and significant performance enhancements [51]. Our implementation comprises the three key layers of the *Lambda* architecture: the batch layer, the speed layer, and the serving layer. The tools selected for our system are based on stringent criteria, including being open-source, scalable, distributed, extensible, and widely validated in production environments. To support this architecture, we adopt a private cloud computing approach. This decision ensures greater control over data security, compliance, and resource allocation while maintaining scalability and flexibility [55]. The private cloud infrastructure is hosted on-premise, serving as the backbone for managing data storage, processing, and analytics. The following sections provide a detailed overview of

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the system's underlying layers and their roles in the framework.

4.1. Master Dataset

Data flowing in the platform goes through several steps. First, data is coming from two main sources (i) dairy farms and (ii) weather station observations. Incoming data can be in various formats, such as XML, JSON, or XLS files received in different ways, such as Web service, Email attachment, or MQTT server, which all depend on the sensor manufacturer being used. Next, data is transmitted to our cloud platform via different network protocols, such as Wi-Fi, 3G/4G, or LoRa. We use EMQ X as an MQTT message broker. Second, when the data arrives at our platform, it will be processed by a data ingestion tool that is responsible for capturing and storing real-time messages to be consumed by a stream/batch processing consumer. In our study, we use Apache Kafka as a data ingestion tool. It can scale to handle millions of messages per second. Third, and after capturing real-time messages, the platform processes them to gain knowledge about the data. Finally, storage tools are used as an output destination to capture real-time and batch data for archiving, or for further processing. In our architecture, we use Apache Cassandra, PostgreSQL, and MinIO for NoSQL, relational data, and unstructured data, respectively. Additionally, PostGIS is used for storing geospatial data.

4.2. Batch Layer

The batch layer is responsible for managing large-scale historical data processing. We utilize Apache Spark, an open-source, distributed processing system, to perform complex computations efficiently. Spark's fault-tolerance and scalability make it ideal for generating accurate, precomputed views of historical data that are periodically updated. This layer processes raw data ingested from diverse sources, such as IoT devices and external systems, and stores the results in a format optimized for querying.

4.3. Speed Layer

The speed layer is designed to process real-time data streams with minimal latency. For this, we use Apache Kafka Streams, a lightweight yet powerful stream-processing library. Kafka Streams processes incoming data in real-time, providing quick and approximate results that are critical for immediate decision-making. This layer ensures a seamless flow of time-sensitive data into the system, complementing the batch layer by focusing on responsiveness and agility.

4.4. Serving Layer

The serving layer is a critical component of the *WALLeSmart* architecture, designed to deliver insights, reporting, and data visualization in real-time. The web application of *WALLeSmart* is structured into backend and frontend

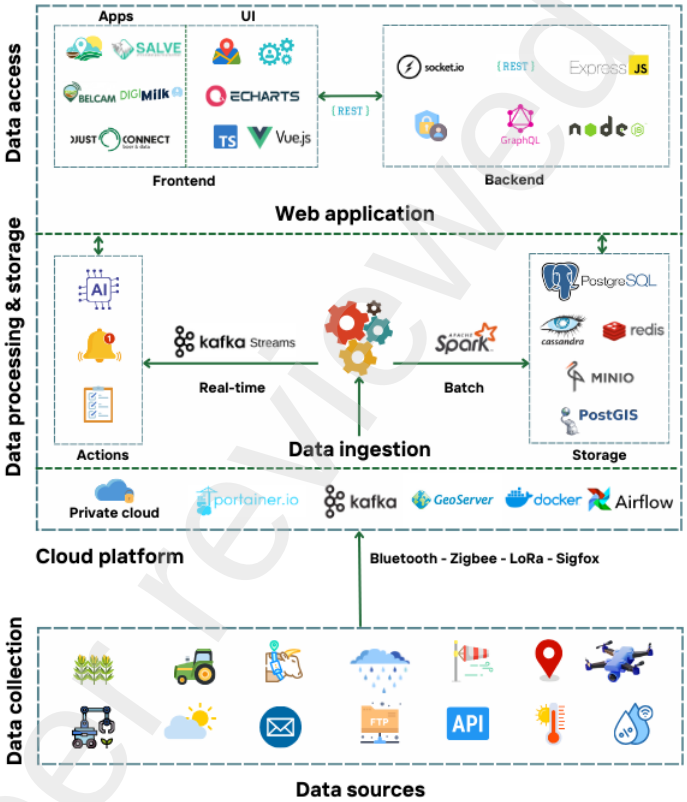

Figure 5: *WALLeSmart* platform architecture.

parts for modularity and scalability. The backend is implemented as a Node.js API server, leveraging Google's V8 JavaScript engine for building fast and scalable network applications. It provides the core functionalities of the platform, including data processing, user authentication, role-based access control, and API endpoints. The frontend is a Vue.js application, ensuring a reactive and user-friendly interface for interacting with the platform. Farm data is visualized through interactive maps powered by OpenStreetMap, while interactive charts are rendered using the Apache ECharts library, offering intuitive and dynamic visualizations. To support researchers and developers, *WALLeSmart* provides a GraphQL Web API for accessing farm and weather data, enabling advanced analysis and modeling. GraphQL's flexible query capabilities allow users to request precise information and deliver a complete description of the data schema. Additionally, GeoServer is integrated into the platform to enable geospatial queries, providing a robust service for accessing spatial data and performing spatial analysis. The architecture of *WALLeSmart* platform is illustrated in Figure 5.

4.5. Development of Decision Support Application in WALLeSmart

The WALLeSmart platform supports the integration of Decision Support Systems (DSS) to enhance its functionality. A standardized development template has been established to streamline the creation of DSSs, ensuring con-

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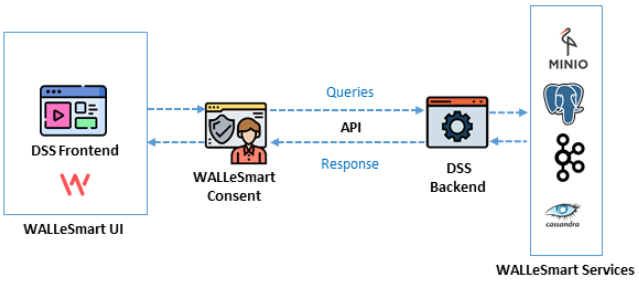

Figure 6: Architecture of a DSS in *WALLeSmart*.

sistency, scalability, and adherence to best practices. Each DSS follows a modular architecture comprising a frontend and a backend. The frontend is developed using Vue.js 3 with TypeScript, Pinia for state management, and Naive UI for customizable user interfaces. The backend, typically implemented in Node.js, handles application logic and communicates with the frontend via HTTP APIs.

Integration into WALLeSmart involves user consent verification managed by the platform before backend interactions. DSSs interface with various services, including MinIO for object storage, PostgreSQL for relational data, Kafka for real-time data processing, and Cassandra for NoSQL storage. The architecture of an OAD within the platform, including its interaction with backend services and the consent management system, is depicted in Figure 6. The development process includes configuring predefined files, testing using modern tools (pnpm, npm), and deploying via containerized solutions (Docker) and automated pipelines (GitLab CI/CD). Once integrated, DSSs are accessible through the WALLeSmart platform, providing seamless interaction for users while leveraging scalable, modular, and secure architecture.

4.6. Connection to External Platforms

The architecture of *WALLeSmart* facilitates secure integration with public and private agricultural platforms that manage user accounts and data. At its core, the system employs secure protocols, including JSON Web Token (JWT) authentication and RSA-based cryptographic key pairs, ensuring secure and efficient communication. The connection process involves user authentication, selecting a platform, and initiating a request by providing an associated email. *WALLeSmart* generates a JWT, sends it to the external platform, and sets the connection status to "pending". The external platform validates the JWT, verifies the email, and updates its status. A confirmation email is sent to the user, and upon approval, the platform notifies *WALLeSmart*, which performs final validations. After verification, the connection status updates to "connected" as shown in the workflow outlined in Figure 7. Once connected, the platforms can securely share user data and enable advanced features such as Single Sign-On (SSO), streamlining access and enhancing functionality for users.

4.7. Security and Privacy

The *WALLeSmart* platform implements multiple layers of security:

- Authentication: Multi-factor authentication using JWT tokens and RSA key pairs
- Authorization: Role-based access control with granular permissions
- Data Privacy: End-to-end encryption for sensitive data transmission
- Audit Logging: Comprehensive logging of system access and changes
- Compliance: Adherence to GDPR and agricultural data privacy regulations

All data exchanges between components are encrypted using TLS 1.3, and sensitive data is stored using AES-256 encryption at rest.

5. Implementation and Results

The WALLeSmart platform is designed to address the needs of the agricultural sector, with a specific focus on the dairy farming sector in the Belgium Wallonia region as a case study. The primary objectives of this case study are to help farmers (i) better manage their dairy cows, including health care, reproduction, milk production, and movement tracking, and (ii) monitor weather conditions, both historical and forecasted, in an intuitive and efficient manner. The data used in this case study is sourced from two main providers: Elevéo[3] for cow behavior data and Agromet[4] for weather data [63].

5.1. Data Sources

Our system integrates data from two primary sources:

1. **Elevéo:** Collected via SenseHub[5] and SmartVel[6] sensors. The dataset spans from January 2007, to February 2025, and includes:
   - 30 dairy farms,
   - 8637 unique cows,
   - 61130 measurement-sets, each consisting of 23 parameters (see Table 3 for details),
   - data comprises 1.5 GB.
2. **Agromet:** Collected from a network of 32 agrometeorological stations covering the entire Wallonia region. The dataset includes:

[3] https://www.awenet.be/awe/commun/asbl/asbl.php
[4] https://agromet.be/
[5] http://www.scrdairy.com/cow-intelligence/sensehub.html
[6] https://www.evolution-xy.fr/fr/monitoring/smartvel

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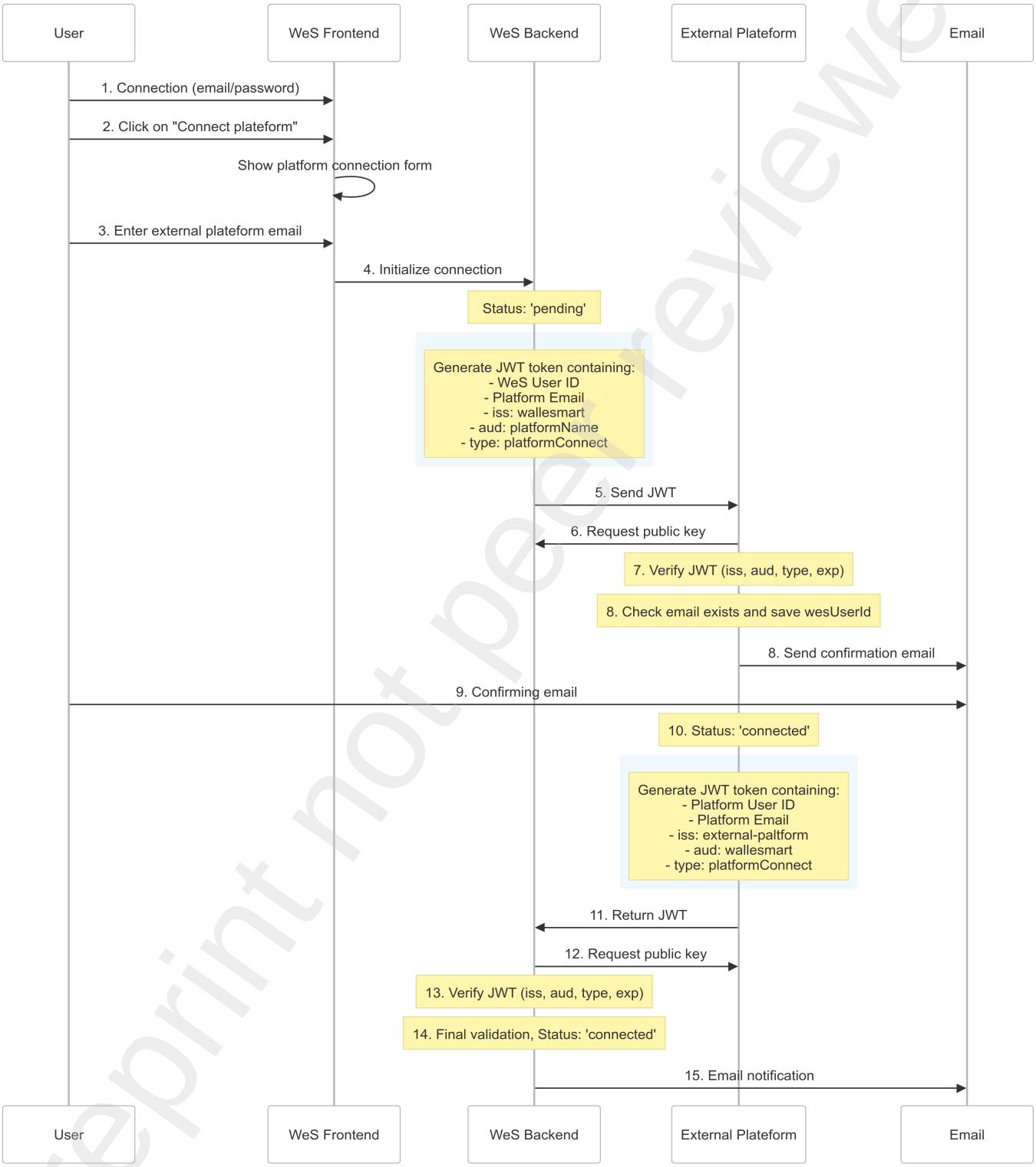

Figure 7: Detailed workflow for secure integration of *WALLeSmart*'s user account with external platforms.

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Table 2: Agromet weather data parameters

Parameter	Description	Unit
tsa	Air Temperature	°C
hra	Relative Humidity	%
tha	Wet Bulb Air Temperature	°C
tsf	Leaf Temperature	°C
tss	Soil Temperature at 20 cm	°C
vvt	Wind Speed at 2 m	m/s
ens	Global Solar Radiation	W/m²
dvt	Wind Direction	°
plu	Precipitation	mm
etp	Reference Evapotranspiration	mm/day
hct	Leaf Wetting Duration	min wet

Table 3: Elevéo milk control data parameters

Parameter variable	Description
BSAUMON	Official animal number
HEURE_TRAIT1_DEB	Controlling date
LAIT_24_VAL, POURC_PROT_24_VAL, POURC_MG_24_VAL	Controlled daily production of milk, protein and fat concentration
CELL_24_VAL, UREE_24_VAL	Controlled cell and urea content
JEL, JET	Lactation days, drying up days
NOLACT	Lactation number
DATE_VEL, DATE_TAR	Date of calving and date of drying up
LAIT, MG, PROT	Cumulative production of milk, protein and fat concentration
LAIT365, MG365, PROT365	Cumulative production over 365 days of milk, protein and fat
PIC	Peak production value

- Historical measurements from January 1, 2008, to January 8, 2025.
- Forecast data for the next 7 days.
- 11 key parameters (see Table 2 for details),
- Over 3 million historical measurements recorded hourly.
- 6879 hourly forecast measurements,
- totaling 3.5 GB of data.

5.2. Deployment

The platform is deployed on a multi-core Linux server, leveraging Docker container technology. This approach isolates the software from its environment, ensuring consistency and uniform functionality across different deployment scenarios. Additionally, *WALLeSmart* adopts a microservice architecture, which decomposes the system into small, independent, and loosely coupled services. Each service is responsible for a specific functionality, such as data ingestion, processing, storage, or visualization.

5.3. The Data Value Chain in WALLeSmart

The WALLeSmart platform follows a well-defined data value chain to ensure efficient data flow and processing. This section describes the key steps in the data value chain, including data acquisition, ingestion, processing, and visualization.

5.3.1. Data Collection

The data acquisition step represents the foundational layer of the platform, responsible for collecting data from various sources. Data is collected from Elevéo and Agromet using SOAP and RESTful APIs on an hourly and daily basis. The platform supports multiple data source formats, including EMQ as the MQTT server for real-time data streams, FTP server for file-based data transfer, and email attachments for manual data uploads. The received data is in XML and JSON formats. Data collection algorithms are implemented in Python scripts, which are scheduled for execution using Apache Airflow for task orchestration and scheduling.

5.3.2. Data Ingestion

In the data ingestion step, the collected data is ingested into the platform for further processing. Python scripts from the data acquisition step act as producers, publishing data to Apache Kafka *topics*. Kafka topics are *partitions* and replicated across multiple servers (*broker*) to ensure scalability and fault tolerance. For MQTT sources, data is published directly from the EMQ server to Kafka topics.

5.3.3. Data Processing

The data processing step involves real-time and batch processing of the ingested data. Data published to Kafka topics is processed using Kafka Streams for real-time stream processing. Kafka Streams applications perform transformations (e.g., date formatting) and collect statistics in real time. For batch processing, Apache Spark is used to handle tasks such as calculating daily cow movement or average temperature. Processed data is republished to Kafka topics and stored in PostgreSQL (for relational data), PostGIS (for geospatial) or Apache Cassandra (for NoSQL data) using Kafka connectors. Task scheduling and workflow management for batch processing are handled by Apache Airflow.

5.3.4. Data Visualization

The final step in the data value chain is data visualization, which provides actionable insights to users through a web application. The application supports multiple user roles:

- Farmers can access Agromet weather forecasts and view a map of the Wallonia region with weather stations and current conditions (see Figure 8b).
- Farmers can efficiently manage their data and application permissions through an intuitive, user-friendly interface. These permissions are associated with various data sources, including Elevéo and ARSIA. The interface features a comprehensive table that clearly displays the status of permissions—such as validated, expired, or pending—across different data categories. Permissions can be managed manually, by the platform, or through a connected web service, offering flexibility and control over data access.

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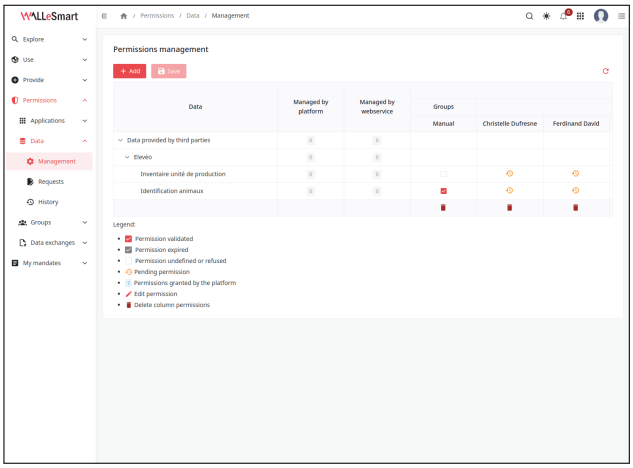
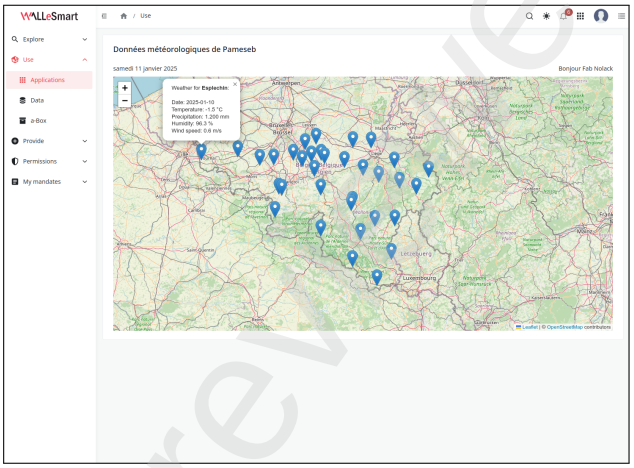
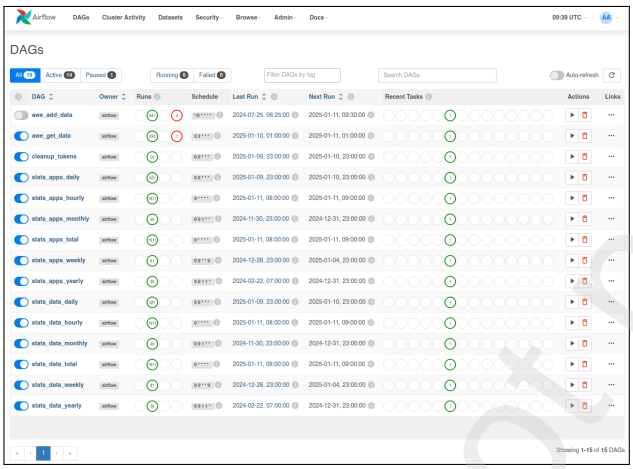


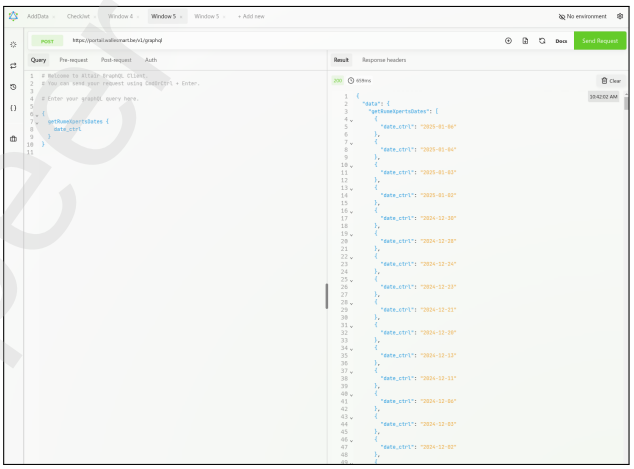
(a) Data permissions management interface in WALLeSmart.



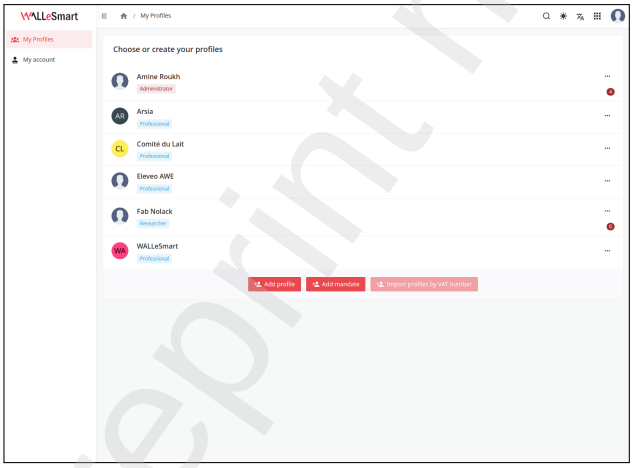
(b) Current weather conditions in Agromet application.



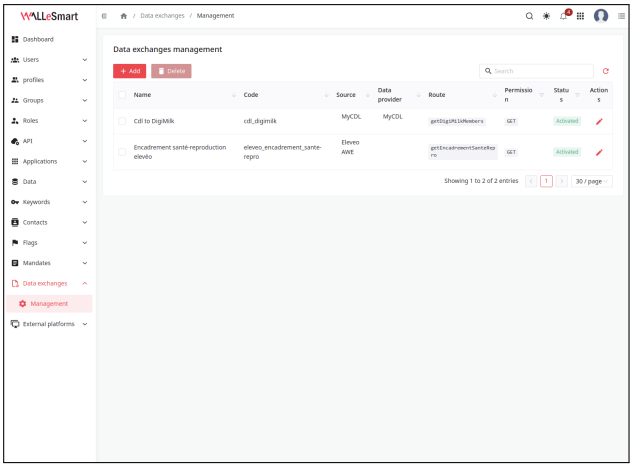
(c) Stream and batch jobs scheduling management in AirFlow.



(d) GraphQL API querying.



(e) Profiles and mandates managements.



(f) Data exchange (API) consent management.

Figure 8: *WALLeSmart* platform and some of its features.

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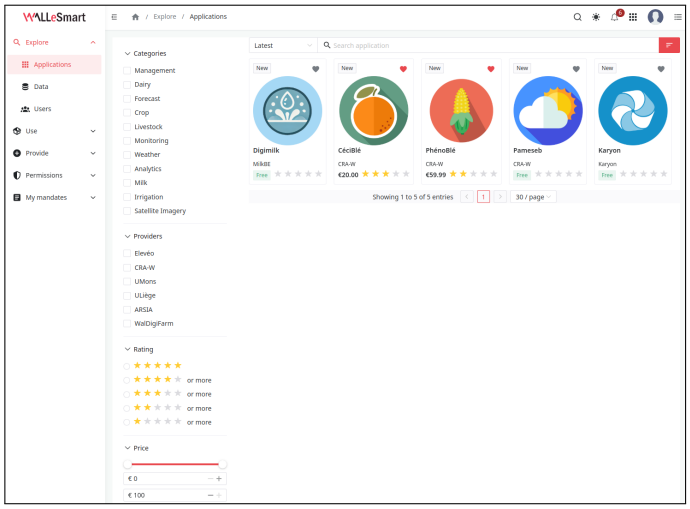

Figure 9: *WALLeSmart* application (DSS) marketplace.

- Administrators can manage stream and batch job scheduling using Apache Airflow (see 8c), edit user profiles, and configure system settings.
- Researchers can query the database using GraphQL API to ask complex questions, such as "*How much milk is produced by lactating cows daily?*" (see Figure 8d).

The web application uses Apache ECharts for interactive charts and integrates with GeoServer for geospatial data visualization. Geospatial data is stored and managed using PostGIS, enabling advanced mapping and spatial analysis.

5.4. The WALLeSmart Features

The *WALLeSmart* platform offers a wide range of features designed to meet the needs of its diverse user base. These features include:

5.4.1. Profiles and Mandates Managements

WALLeSmart provides robust user profile and mandate management capabilities. Users can create and manage profiles with specific roles and permissions, ensuring that access to data and features is tailored to individual responsibilities. This feature supports multiple user roles, such as farmers, administrators, and researchers, each with customized access levels and functionalities. Central to the system's functionality is a mandate mechanism that enables delegated access through carefully defined parameters, including principal and representative designations, feature-specific permissions, and temporal constraints. The profile and mandats management interface is intuitive and user-friendly, as shown in Figure 8e. Additionally, the platform can automatically import user profiles using their VAT numbers, streamlining the onboarding process and reducing manual effort.

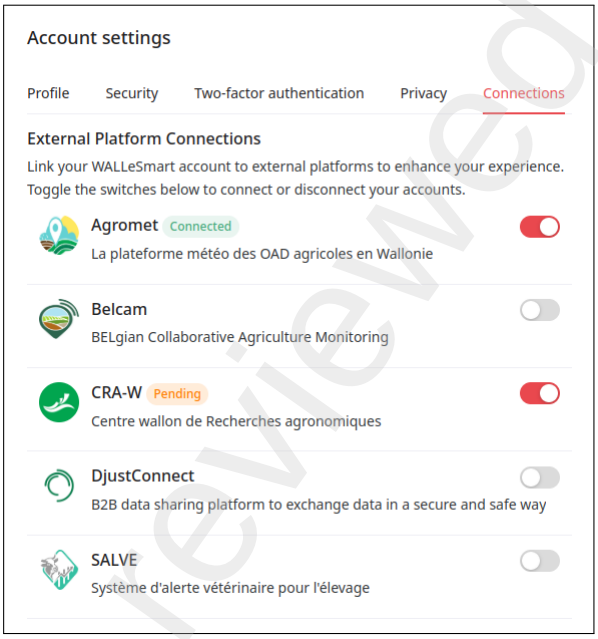

Figure 10: External platforms management in *WALLeSmart*.

5.4.2. WALLeSmart Application Marketplace

The *WALLeSmart* platform features a well-organized application marketplace that showcases various DSSs (see Figure 9). The interface displays a clean layout with several key components, such as a navigation menu with filtering options including categories, providers, rating, and price range. Before accessing an application, users must submit a request for access. This request is reviewed by the application's developer, who can either approve or reject it based on specific criteria, such as compliance with licensing agreements. Once permission is granted, users can access the application through the "Use" menu in the *WALLeSmart* platform. This workflow ensures secure, responsible usage of applications while maintaining a user-friendly experience.

5.4.3. Data Exchange Consent

Figure 8f illustrates the *WALLeSmart* data exchange management interface, designed to facilitate secure and controlled data sharing between the platform and third-party systems. The interface enables administrators to manage data exchange configurations. For instance, APIs such as "getMilkControl", belonging to third-party partners like RumeXperts, can be integrated to enable seamless data transfer. Users have the ability to approve or deny third-party access to their data via the *WALLeSmart* platform, ensuring transparency and user consent in all data-sharing activities.

5.4.4. External Platforms Connection

Figure 10 illustrates the interface for managing external platform connections within the *WALLeSmart* account settings. This feature allows users to link their account

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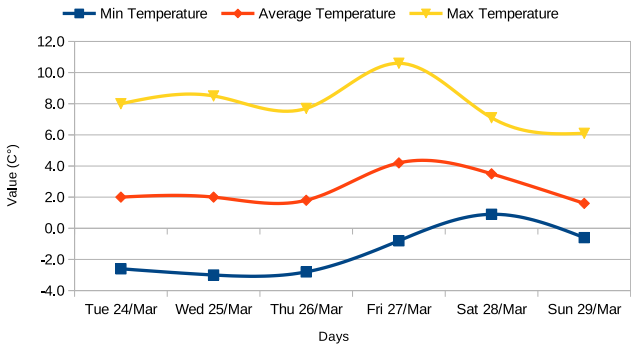
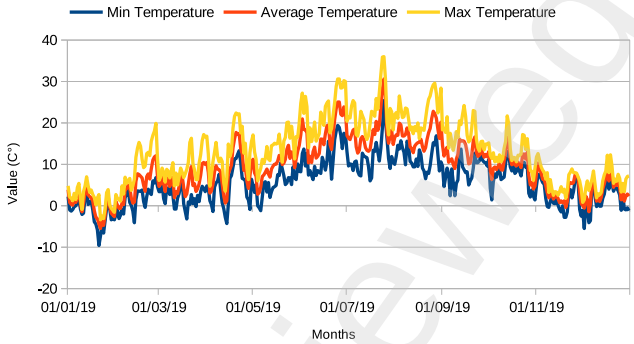
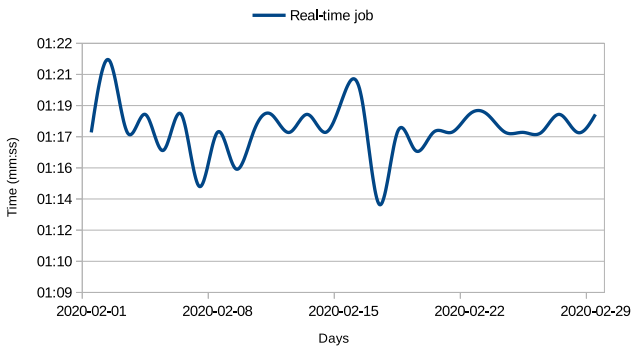


(a) Real-time job for calculating temperature forecasting for the next 7 days.

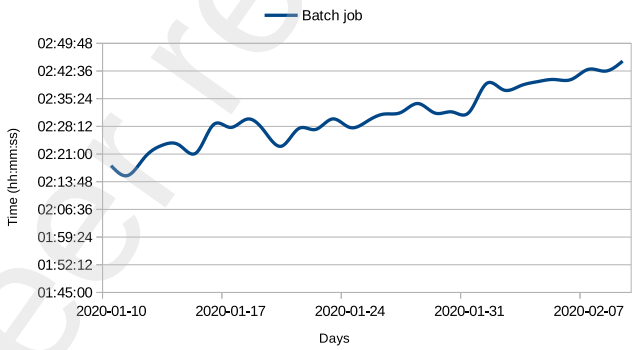


(b) Batch job for calculating temperature history for 2019.

Figure 11: Examples of batch and real-time jobs for scheduled one weather station.



(a) The execution time of a real-time job for weather data over one month.



(b) The execution time of a batch job for dairy data over one month.

Figure 12: Performance of batch and real-time job execution.

to a range of public and private platforms that manage farmer data. Platforms such as Agromet, Belcam, Djust-Connect, and SALVE can be connected to enable secure data sharing and advanced integration.

Those results are shown in the speed layer through a user-friendly Web interface. This can help farmers having a closer look at weather data and timely decision making, for example, to take an estimation of irrigation needs and adjust irrigation schedules accordingly.

5.5. Performance Evaluation

This section presents the experiments and results conducted to evaluate the effectiveness of our platform. We focus on both real-time and batch processing tasks, highlighting their performance and outcomes.

5.5.1. Real-Time Job Execution

Figure 11a illustrates the results of a real-time job executed by the speed layer. The task involves calculating the average, maximum, and minimum daily temperatures for the next seven days. Since the platform receives estimated temperature data on an hourly basis, the real-time layer aggregates this data to produce daily forecasts. The data stream is ingested into Kafka topics and processed using Kafka Streams, which performs transformations and computations in real time. The results are then republished to Kafka and stored in Cassandra. This pipeline ensures efficient and timely processing of real-time data.

5.5.2. Batch Job Execution

Figure 11b showcases the results of a batch job executed by the batch layer. The task involves calculating the average, maximum, and minimum daily temperatures for the entire year of 2019 across all weather stations. This batch process is scheduled to run at regular intervals and operates on the entire master dataset. Leveraging Apache Spark, the batch layer accelerates computation, and the results are saved to the database upon completion.

5.5.3. Execution Time Analysis

To further evaluate the performance of our architecture, we measured the execution times for both real-time and batch jobs over an extended period.

- **Real-Time Job Execution Time:** Figure 12a shows the execution time of the real-time job over a one-month period. The average execution time is consistently low, averaging 1 minute and 18 seconds per

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job. This efficiency is attributed to the lightweight and scalable nature of Apache Kafka Streams, which processes data streams with minimal latency. The reported time includes data ingestion, processing, and storage.

- **Batch Job Execution Time:** Figure 12b depicts the execution time of a batch job analyzing dairy farm data. The average execution time is significantly longer, averaging 2 hours and 31 minutes. The curve exhibits an upward trend over time, reflecting the increasing volume of dairy data processed daily. To address this, the system can be scaled horizontally by adding more server nodes to distribute the computational load and improve performance.

These experiments demonstrate the platform's ability to handle both real-time and batch processing tasks effectively. While real-time jobs exhibit low and consistent execution times, batch jobs require optimization for scalability as data volumes grow. Future improvements include parallelizing computations and enhancing resource allocation to further reduce batch job execution times.

5.6. Decision Support Systems

In this section, we highlight several Decision Support Systems (DSS) that leverage our platform ecosystem.

5.6.1. SALVE

SALVE[7] is a web-based Veterinary Alert System that automates livestock data analysis to enhance decision-making and reduce reaction times. It collects data from around 100 dairy and meat farms in Wallonia, including herd composition, milk quality, and climatic conditions. By integrating *WALLeSmart*'s real-time milk data, *SALVE* analyzes health, productivity, and environmental risks using advanced algorithms. This enables early detection of issues, alerting farmers 30 to 45 days before problems become critical. As a result, *SALVE* improves animal welfare, extends livestock longevity, and reduces production costs.

5.6.2. W@llHerbe

W@llHerbe is an innovative application designed to continuously track grass growth by integrating field measurements, satellite imagery, and growth model data. This tool provides accurate estimates of both the quantity and quality of grass available in pastures. It will be seamlessly integrated into the *WALLeSmart* app store, ensuring easy access for users, and will leverage WALLeSmart data to enhance its capabilities. Paired with a digitized grazing calendar, *W@llHerbe* empowers farmers to optimize plot management, schedule mowing more effectively, and refine livestock rations. This results in better utilization of forage resources, tailored to the challenges of today's changing climate.

5.6.3. DigiMilk

DigiMilk[8] is a new digital tool launched by MilkBE and hosted on the *WALLeSmart* aplatform as part of the updated Sustainability Monitor for Belgium's dairy sector. It empowers dairy farmers to easily access, update, and manage their sustainability data, encompassing over 70 initiatives across nine key areas, including animal welfare and climate. By integrating automatic data collection from existing databases—such as those of the Wallonia Government, Arsia, and CdL—*DigiMilk* streamlines administrative tasks, allowing farmers to input updates at their convenience. With explicit consent through *WALLeSmart* platform, farmers can securely share their data with MilkBE or dairy processors, contributing to sector-wide sustainability goals while retaining full control over their information.

5.7. Limitations

The *WALLeSmart* platform, while pivotal for advancing sustainability in Wallonia's agricultural sector, faces several limitations that hinder its adoption and performance. These challenges can be grouped into operational constraints at the farm level and technical hurdles within the platform itself.

5.7.1. Operational constraints

Slow connectivity at farm offices, a common issue in rural areas, delays data uploads and interactions with the platform, frustrating users who rely on timely updates. Similarly, the absence of power in fields limits the deployment of IoT devices essential for real-time data collection, leaving gaps in on-site monitoring. This problem is compounded by a lack of connectivity in fields, where poor or nonexistent network coverage prevents seamless data transmission from sensors. Additionally, precision mapping suffers due to limited sensor availability, restricting the platform's ability to generate detailed, accurate sustainability metrics—such as soil health or emissions—that farmers need to meet sector-wide goals.

5.7.2. Technical limitations

Technical limitations further challenge *WALLeSmart*'s performance as it integrates new data sources and scales across the region. Combining diverse datasets, such as weather patterns and dairy farm records, requires manual calibration to align heterogeneous data formats, slowing processing times and introducing potential inconsistencies. Scalability poses another concern, as batch processing performance declines when the number of connected IoT nodes increases, reducing efficiency as adoption grows. Additionally, the platform's plugin system, though versatile, presents user adoption barriers due to its complexity, demanding a level of technical proficiency that many farmers lack and requiring substantial training to ensure effective utilization.

[7] `https://www.salve.vet/`

[8] `https://www.digimilk.be/`

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6. Conclusions and Future Work

Agriculture remains a vital sector, directly impacting global food security and environmental sustainability. As agricultural challenges intensify due to climate variability, resource constraints, and increasing demand, the ability to manage complex agricultural ecosystems efficiently has never been more critical. The richness of big data generated by emerging agri-technologies offers unprecedented opportunities to address these challenges, ensuring a secure and healthy future for both people and the planet.

In this paper, we introduced *WALLeSmart*, a novel cloud-based smart farming management platform designed to harness the power of big data and cloud computing in agriculture. We began by investigating state-of-the-art platforms, big data and cloud architecture initiatives, which informed the design of our solution. Drawing inspiration from the *Lambda* architecture, we proposed a robust framework for the acquisition, processing, storage, and visualization of real-time agricultural data. Our platform integrates advanced technologies for real-time stream processing, such as Apache Kafka Streams, and batch processing using Apache Spark, ensuring scalability and efficiency.

An initial prototype of the platform was developed and tested with 30 dairy farms and 32 weather stations in the Wallonia region of Belgium. The backend demonstrated robust data management capabilities, while the frontend received positive feedback for its user-friendly interface and intuitive design. We also presented examples of batch and real-time job execution, showcasing the platform's ability to handle diverse data processing tasks effectively.

While this study presents a scalable smart farming solution, several promising directions for future research remain. We plan to develop and integrate more decision-making applications into the platform, further enhancing its ability to provide actionable insights to farmers. The development of mobile applications for on-the-go access is also in the pipeline. Integrating advanced AI/ML techniques will further enhance predictive analytics and optimize resource management. Similarly, combining edge computing with cloud systems could enable real-time decision-making, crucial for time-sensitive operations. Further exploration of geospatial analytics, incorporating satellite and drone imagery, may yield more precise agricultural insights. Additionally, establishing interoperability frameworks and standardizing data exchange protocols can foster greater collaboration across platforms. Finally, advancing climate-resilient farming systems is critical to tackling the challenges posed by climate change and ensuring agricultural sustainability. These avenues highlight the immense potential for further innovation in smart farming systems to support global food security and environmental stewardship.

Declaration of generative AI and AI-assisted technologies in the writing process

In the preparation of this work, the authors utilized ChatGPT 3.5 (OpenAI, USA) to assist with grammar refinement. Following its use, the authors thoroughly reviewed and revised the content to ensure accuracy, clarity, and alignment with the intended message. The authors take full responsibility for the final content and conclusions presented in this publication.

Funding

This work is supported by the Wallonia region of Belgium (SPW) under grant DGO3-F114PP011, project 2019-2025.

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